19/11/2025

Phase-4_Code-D

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720

NCERT Booster Test Series for NEET-2026_RM(P4)_NBTS-01D

Time : 180 Min.

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics
Botany: Cell: The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification, Morphology of Flowering Plants
Zoology: Structural Organisation in Animals–Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation, Excretory Products & their Elimination, Locomotion & Movement

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. *A* : Dimensionally different quantities may be added.
R : Dimensionless quantities are necessarily unitless.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

2. A packet contains silver powder of mass 20.23 g ± 0.01 g. Some of the powder of mass 5.75 g ± 0.01 g is taken out from it. The mass of the powder left back is

(1) 14.48 g ± 0.00 g

(2) 14.48 g ± 0.02 g

(3) 14.5 g ± 0.1 g

(4) 14.5 g ± 0.2 g

3. The 9th division of vernier scale coincides with 6th division of main scale in a vernier calliper. If one main scale division is 0.01 mm, then the least count of instrument is

(1) 0.01 mm

(2) 0.001 mm

(3) 0.0033 mm

(4) 0.0066 mm

4. If the unit of force is 10 kN, length is 10 km and time is 1000 s, then the unit of mass will be

(1) 10^3 kg

(2) 10^4 kg

(3) 10^5 kg

(4) 10^6 kg

5. A physical quantity Z is given by $Z = \frac{\sqrt[3]{A}\sqrt[3]{B}}{\sqrt[3]{C}}$ and percentage error in A, B and C are 3%, 4% and 5% respectively. The percentage error in Z is

(1) 4%

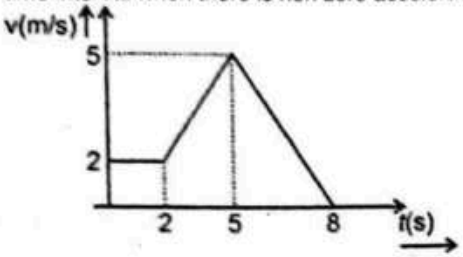
(2) 1%

(3) 2%

(4) 3%

CC-044

6. Velocity-time (v-t) graph for a moving object is shown in the figure. The net displacement of the object, during the time interval when there is non zero acceleration, will be



(1) 9 m

(2) 18 m

(3) 6 m

(4) 12 m

7. A person travels along a straight road for the first $\frac{t}{3}$ time with a speed v_1 and for next $\frac{2t}{3}$ time with a speed v_2. Then the mean speed v is given by

(1) $v = \frac{v_1 + 2v_2}{3}$

(2) $\frac{1}{v} = \frac{1}{3v_1} + \frac{2}{3v_2}$

(3) $v = \frac{1}{3}\sqrt{2v_1 v_2}$

(4) $v = \sqrt{\frac{3v_1}{2v_2}}$

8. A ball A is thrown up vertically with a speed u and at the same instant another ball B is released from a height h. At time t $\left(t < \sqrt{\frac{2h}{g}}\right)$, speed of A relative to B is

(1) u

(2) $2u$

(3) $u - gt$

(4) $\sqrt{u^2 - gt}$

9. A ball is thrown vertically upwards from the ground. It crosses a point at height of 60 m twice at an interval of 4 s. The velocity with which ball will hit the ground is

(1) 40 m/s

(2) 20 m/s

(3) 30 m/s

(4) 15 m/s

10. Velocity v and position x of a body are related as $v^2 = Kx$, where K is a constant. Velocity of the body after 1 s is (given that at $t = 0$, $x = 0$)

(1) K

(2) $K/2$

(3) $2K$

(4) $\sqrt{K}$

11. A : The angle between velocity vector and acceleration vector in uniform circular motion is 90°.
R : When particle increases its speed in a circular motion, acceleration still remains towards centre.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

12. A person walking at 4 m/s, finds rain drops falling on his head vertically with speed of 4 m/s. The actual speed of rain drop is

(1) 6 m/s

(2) 4 m/s

(3) $4\sqrt{2}$ m/s

(4) 10 m/s

13. A river is flowing due east with a speed 2 m/s. A swimmer on the south bank of river can swim in still water at a speed of 4 m/s. Swimmer swims in such a way that he reaches a directly opposite point on the north bank. The resultant speed of the swimmer is

(1) 5 m/s

(2) $2\sqrt{3}$ m/s

(3) $3\sqrt{2}$ m/s

(4) 4 m/s

14. A hose pipe on ground shoots a stream of water upwards at an angle of 60° to horizontal with a velocity of 10 ms^{-1}. The height at which the water will strike a wall 5 m away, is

(1) 2.46 m

(2) 3.76 m

(3) 4.9 m

(4) 8.66 m

15. A projectile is given an initial velocity of $(2\hat{i} + 3\hat{j})$ m/s, where $\hat{i}$ is along horizontal and $\hat{j}$ is along the vertical. The equation of trajectory of projectile is (g = 10 m/s^2)

(1) $4y = 3x^2 - 8x$

(2) $y = 12x + 10x^2$

(3) $8y - 10x^2 + 12x = 0$

(4) $10x^2 - 12x + 8y = 0$

16. An unbanked road has a radius of curvature equal to 30 m. The maximum speed at which a car can make a turn if the coefficient of friction is 0.75, is

(1) 10 m/s

(2) 30 m/s

(3) 15 m/s

(4) 5 m/s

17. A machine gun fires 10 bullets per second with speed 10 m/s. If mass of each bullet is 300 gm, then the force required to keep the gun stationary is

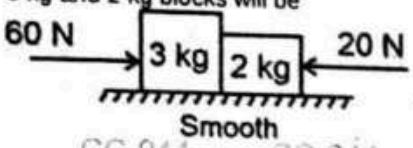
(1) 40 N

(2) 10 N

(3) 20 N

(4) 30 N

18. In the diagram shown, the normal reaction force between 3 kg and 2 kg blocks will be



CC-044 CC-044 CC-044 CC-044

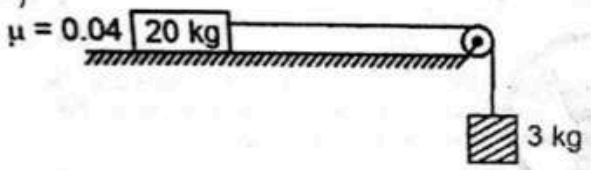
(1) 24 N

(2) 30 N

(3) 36 N

(4) 48 N

19. What is the acceleration of block placed on rough surface. The mass of block is 20 kg and is connected by light string to hanging mass of 3 kg as shown in figure. (g = 10 m s^{-2})



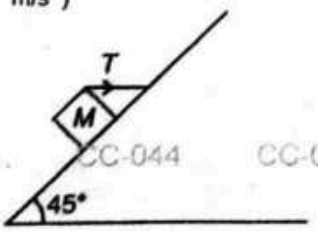
(1) $\frac{15}{19}$ m s^{-2}

(2) $\frac{21}{22} ms^{-2}$

(3) $\frac{22}{23} ms^{-2}$

(4) $\frac{15}{19}$ m s^{-2}

20. A small block of mass 15 kg is just kept at rest on a rough inclined plane as shown in figure, with the help of a horizontal string which has a tension of 50 N. The coefficient of friction between surface of contacts is (g = 10 m/s^2)



CC-044 CC-044 CC-044 CC-044

(1) $\frac{1}{2}$

(2) $\frac{2}{3}$

(3) $\frac{3}{4}$

(4) $\frac{1}{4}$

21. When a spring is elongated by a cm, the potential energy stored is 2 J. If this spring is further elongated by a cm, the increase in potential energy will be

(1) 2 J

(2) 4 J

(3) 3 J

(4) 6 J

22. A ball of mass m is dropped from a height h from the ground. It bounces back with speed $\sqrt{gh}$ m/s. The coefficient of restitution between the ball and ground is

(1) $\sqrt{2}$

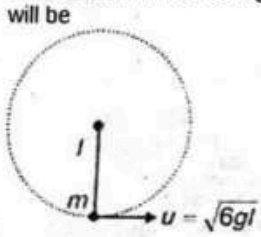
(2) $\frac{1}{\sqrt{2}}$

CC-044 CC-044 CC-044

(3) 1

(4) Zero

23. In a vertical circular motion as shown in figure, if particle of mass m is imparted a speed $\sqrt{6gl}$ at the lowest point, then tension in the string of length l at the topmost point will be



(1) mg

(2) $\frac{mg}{2}$

(3) $\frac{3}{2}mg$

(4) $2mg$

24. A body of mass m is projected with velocity u, at an angle θ from horizontal with the ground. When it is at the maximum height, then instantaneous power delivered by gravitational force will be

(1) Zero

(2) $\frac{mgu\cot\theta}{\sqrt{2}}$

(3) $\frac{mgu\tan\theta}{\sqrt{2}}$

(4) $\frac{mgu}{2\sqrt{2}}$

CC-044 CC-044 CC-044

25. A body of mass 2 kg moving with a velocity of 3 m/s collides head on with a body of mass 1 kg moving with a velocity of 4 m/s in opposite direction. After collision the two bodies stick together and move with a common velocity whose magnitude is equal to

(1) $\frac{1}{3}$m/s

(2) $\frac{2}{3}$m/s

(3) $\frac{1}{4}$m/s

(4) $\frac{3}{4}$m/s

26. A boy of mass 30 kg is standing at one end on a stationary boat over a lake. The boat has a mass 70 kg and length 10 m. Now the boy walked to the other end of the boat. Relative to the ground

(1) Boy has moved by a distance $\frac{40}{7}$ m

(2) Boy has moved by a distance $\frac{30}{7}$ m

(3) Boat has moved by a distance 7 m

(4) Boat has moved by a distance 3 m

27. A rigid body can be hinged about any point on the x-axis. When it is hinged such that the hinge is at x, the moment of inertia is given by $I = x^2 - 24x + 48$. The x-coordinate of centre of mass is

(1) 6

(2) 12

(3) 24

(4) 4

28. Match the Column I with Column II.

	Column I		Column II
(A)	Moment of inertia of rod (m, l) about its one end.	(P)	$\frac{ml^2}{8}$
(B)	Moment of inertial of hollow sphere (mass-m, Dia-l) about its diameter	(Q)	$\frac{2}{3}ml^2$
(C)	Moment inertia of square plate (mass m, side length l) about axis passing through its centre and perpendicular to its plane	(R)	$\frac{ml^2}{6}$
(D)	Moment of inertia of cylinder (mass-m, Dia-l) about its axis	(S)	$\frac{ml^2}{3}$

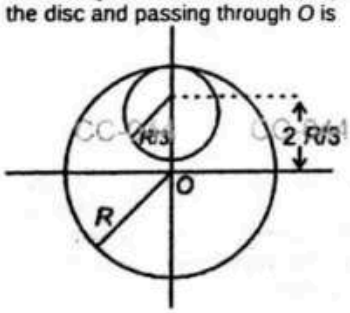
(1) A(P); B(Q); C(R); D(S)

(2) A(S); B(Q); C(S); D(R)

(3) A(S); B(R); C(Q); D(P)

(4) A(S); B(R); C(R); D(P)

29. From a circular disc of Radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is



(1) $4MR^2$

(2) $\frac{40}{9}MR^2$

(3) $10MR^2$

(4) $\frac{37}{9}MR^2$

30. If the polar ice caps of the earth melts, then the length of day will

(1) Increase

(2) Decrease

(3) Remains same

(4) Earth stops to rotate

31. The time period of a satellite in a circular orbit of radius R is T. The period of another satellite in a circular orbit of radius $6R$ for same planet will be

(1) $4T$

(2) $6\sqrt{6}T$

(3) $\frac{T}{6}$

(4) $\frac{T}{\sqrt{6}}$

32. The height at which the weight of a body becomes $\frac{1}{4}$th of its weight on the surface of earth (Radius R), is

(1) $\frac{3R}{5}$

(2) $\frac{R}{4}$

(3) R

(4) $2R$

33. The escape velocity of a body from earth surface is V. The escape velocity of same body from a height of $7R$ from earth surface is (R is radius of earth)

(1) $\sqrt{2}V$

(2) $\frac{V}{\sqrt{2}}$

(3) $\frac{V}{2}$

(4) $\frac{V}{2\sqrt{2}}$

34. The potential energy of a system of four particles each of mass m, placed at vertices of a square of side a is

(1) $-(4+\sqrt{2})\frac{Gm^2}{a}$

(2) $-4\sqrt{2}\frac{Gm^2}{a}$

(3) $-\frac{4Gm}{a}$

(4) $-4\frac{Gm^2}{a}$

35. A satellite of mass m is orbiting in a circular path of radius r around the earth. If its kinetic energy is E, then its angular momentum is

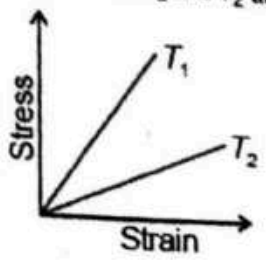
(1) $\left(\frac{2}{3}mrE\right)^{1/2}$

(2) $(2mr^2E)^{1/2}$

(3) $(mr^2E)^2$

(4) $\left(\frac{1}{2}mrE\right)^{1/2}$

If stress versus strain plot for a metal at two different temperatures T_1 and T_2 are shown below, then



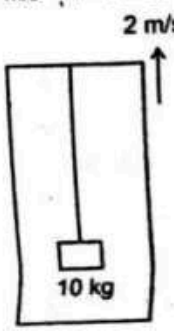
(1) $T_1 > T_2$

(2) $T_1 < T_2$

(3) $T_1 = T_2$

(4) $T_1 = \frac{1}{T_2}$

37. One end of a light steel wire is fixed to ceiling of an elevator moving up with an acceleration of 2 m/s^2 and a load of 10 kg hangs from other end. If cross sectional area of wire is 2 mm^2, the longitudinal strain in wire is (g = 10 m/s^2, $Y = 2 \times 10^{11}$ N/m^2)



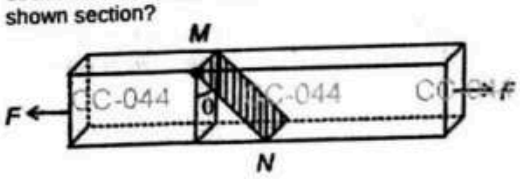
(1) 2.5×10^{-5}

(2) 3.0×10^{-4}

(3) 2.0×10^{-5}

(4) 2.5×10^{-4}

38. A rod, shown in figure, is subjected to equal and opposite forces. MN is a plane making an angle $\theta = 45°$ with cross-section A of the rod. What is tensile stress on MN shaded shown section?



(1) $\frac{\sqrt{2}F}{A}$

(2) $\frac{F}{2A}$

(3) $\frac{F}{A}$

(4) $\frac{2F}{A}$

39. Force constant of a spring (k) is analogous to (symbols have their usual meaning)

(1) $\frac{YA}{L}$

(2) $\frac{YL}{A}$

(3) $\frac{AL}{Y}$

(4) ALY

40. Young's modulus of the material of a wire of length L and radius r is Y. If the length of the wire is reduced to $\frac{L}{4}$ and radius is reduced to $\frac{r}{2}$, then Young's modulus

(1) Will be $2Y$

(2) Will be $Y/2$

(3) Will be $4Y$

(4) Will remain Y

41. What is the total pressure on a swimmer 20 m below the surface of a lake? (Atmospheric Pressure : 10^5 Pa)

(1) 2×10^5 Pa

(2) 1×10^5 Pa

(3) 4×10^5 Pa

(4) 3×10^5 Pa

42. Hydraulic lift works on the principle of

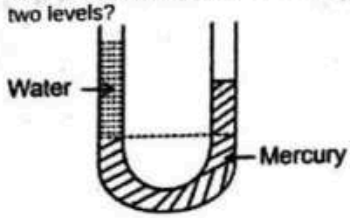
(1) Stokes' law

(2) Pascal's law

(3) Archimedes' principle

(4) Bernoulli's principle

43. A U tube open at both ends contains mercury. 15 cm of water is added to left arm. What is difference of height of two levels?



(1) 7 cm

(2) 13.9 cm

(3) 8 cm

(4) 4.1 cm

CC-044

44. If the angle of contact θ in a capillary is more than 90° then the level of liquid in the capillary

(1) Rises

(2) Drops

(3) Remains same

(4) Fluctuates

45. Eight small raindrops each of radius 1.5 mm, are falling down with a terminal velocity of 8 m/s. They coalesce to form a bigger drop. The terminal velocity of the bigger drop is

(1) 16 m/s

(2) 32 m/s

(3) 64 m/s

(4) $16\sqrt{2}$ m/s

CC-044 CC-044 CC-044 CC-044 CC-044 CC-044 CC-044

CHEMISTRY

46. Which of the following has maximum number of electrons?

(1) 72 g of O_3

(2) 34 g of NH_3

(3) 1.5 g molecules of SO_2

(4) 44.8 L of CO_2 at STP

47. 4 g of Mg is reacted with 8 g of O_2. Mole of MgO produced in the reaction is

(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) $\frac{1}{6}$

(4) $\frac{1}{3}$

48. Numerical prefix used for 10^{12} is

(1) Giga

(2) Tera

(3) Peta

(4) Exa

CC-044 CC-044 CC-044 CC-044 CC-044 CC-044

49. Number of Fe atoms in 100 g Haemoglobin if it contains 0.33% Fe. (Atomic mass of Fe = 56)

(1) 0.035×10^{23}

(2) 35

(3) 3.5×10^{23}

(4) 7×10^{8}

50. To which species, Bohr's theory is not applicable?

(1) Li^{2+}

(2) B^{4+}

(3) He^{+}

(4) H^{+}

51. If mole fraction of solvent in aq. H_2SO_4 solution is 0.85 then number of moles of solute present in 2 kg of solvent is

(1) 6.66

(2) 5.66

(3) 0.15

(4) 19.6

Match the given columns

Column I (Composition)		Column II (Molality)
a.	18 g glucose ($C_6H_{12}O_6$) in 180 g water	(i) $\frac{25}{9}\ m$
b.	34.2 g sucrose ($C_{12}H_{22}O_{11}$) in 360 g water	(ii) $\frac{50}{9}\ m$
c.	5.85 g of sodium chloride (NaCl) in 18 g of water	(iii) $\frac{5}{18}\ m$
d.	4.6 g ethyl alcohol (C_2H_5OH) in 36 g water	(iv) $\frac{5}{9}\ m$

Choose the correct option.

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iv), b(iii), c(i), d(ii)

53. All of the following spectral series are found in infrared region **except**

(1) Lyman

(2) Paschen

(3) Pfund

(4) Brackett

54. Given below are two statements:

Statement I: Equal masses of O_2 and N_2 contain equal number of atoms.

Statement II: Molality and mole fraction are temperature independent quantities.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

55. In an isothermal process, entropy change for 1 mol of substance is expressed as

(1) $\Delta S = \frac{q_{rev}}{T^2}$

(2) $\Delta S = 2R \ln \frac{V_1}{V_2}$

(3) $\Delta S = nRT \ln \left(\frac{P_1}{P_2}\right)$

(4) $\Delta S = R \ln \left(\frac{V_2}{V_1}\right)$

56. Match list I with list II

List I (Quantum numbers)		List II (Orbital)
a.	$n = 2, \ell = 0$	(i) 2p
b.	$n = 3, \ell = 1$	(ii) 3d
c.	$n = 3, \ell = 2$	(iii) 3p
d.	$n = 2, \ell = 1$	(iv) 2s

Choose the correct answer from the options given below:

(1) a–(iii), b–(iv), c–(ii), d–(i)

(2) a–(ii), b–(iii), c–(iv), d–(i)

(3) a–(iv), b–(i), c–(ii), d–(iii)

(4) a–(iv), b–(iii), c–(ii), d–(i)

57. If a process is spontaneous at only high temperature, then

(1) $\Delta H < 0$ and $\Delta S > 0$

(2) $\Delta H > 0$ and $\Delta S > 0$

(3) $\Delta H > 0$ and $\Delta S < 0$

(4) $\Delta H < 0$ and $\Delta S < 0$

58. The difference between heat of reaction at constant pressure and constant volume for the reaction $2C_6H_6(l) + 15O_2(g) \rightarrow 12CO_2(g) + 6H_2O(l)$ at 25°C in kJ is

(1) +7.43

(2) +4.83

(3) –7.43

(4) –4.83

59. Intensive property among the following is

(1) Volume

(2) Number of mole

(3) Pressure

(4) Surface area

60. Which of the following reactions represents standard enthalpy of formation of HBr(g)?

(1) $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$

(2) $\frac{1}{2}H_2(g) + \frac{1}{2}Br_2(g) \rightarrow HBr(g)$

(3) $\frac{1}{2}H_2(g) + \frac{1}{2}Br_2(l) \rightarrow HBr(g)$

(4) $H_2(g) + Br_2(l) \rightarrow 2HBr(g)$

61. For chlorine atom, total number of electrons which have $(n+l)$ value equal to three is

(1) 2

(2) 6

(3) 8

(4) 4

62. Which among the following are state functions?
(a) q
(b) w
(c) q + w
(d) ΔH
(e) ΔS

(1) (a) and (b) only

(2) (a), (b) and (c) only

(3) (c), (d) and (e) only

(4) (b), (d) and (e) only

63. The correct order of wavelength of given electromagnetic radiations is

(1) Radio waves > Microwaves > IR > Visible

(2) Microwaves > Radio waves > IR > Visible

(3) Visible > IR > Microwaves > Radio waves

(4) IR > Visible > Microwaves > Radio waves

64. A gas is allowed to expand in an insulated container against a constant external pressure of 4.0 atm from an initial volume of 0.75 L to a final volume of 6.75 L. The change in internal energy of the gas will be

(1) –2.43 kJ

(2) –3.21 kJ

(3) –1.32 kJ

(4) –5.45 kJ

65. Consider the given statements:
Statement I : If enthalpy of atomisation of methane 1665 kJ mol^{-1} then mean bond enthalpy of C – H bond is 416.25 kJ mol^{-1}.
Statement II : In methane, all the four C – H bonds are equal in bond length and the energies required to break the individual C – H bonds in each successive step will also be the same.
In the light of above statements choose the correct option.

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

66. In a process, 600 J of heat is absorbed by the system and 250 J of work is done by the system, change is internal energy during the process is

(1) 100 J

(2) 850 J

(3) 350 J

(4) 250 J

67. For the reaction,
$3A(g) + B(s) \rightarrow 3C(g)$
If $\Delta U^\circ = -10$ kJ, $\Delta S^\circ = -50\ JK^{-1}$
Calculate the value of ΔG° at 298 K

(1) –4.9 kJ

(2) –14.9 kJ

(3) 4.9 kJ

(4) 14.9 kJ

68. Standard molar enthalpy of formation of which pair of substances at 298 K has zero value?

(1) $N_2(g)$ and $S_{rhombic}(s)$

(2) $I_2(s)$ and $P_4(g)$

(3) $Br_2(\ell)$ and $HF(g)$

(4) $C_{diamond}(s)$ and $Cl_2(g)$

69. The correct order of decreasing bond length of C–C, C–H, C–O and O–H is

(1) C–C > C–O > C–H > O–H

(2) C–O > C–C > C–H > O–H

(3) C–O > C–C > O–H > C–H

(4) C–C > C–O > O–H > C–H

70. Which of the following have non-zero net dipole moment?
a. BeF_2
b. BF_3
c. H_2O
d. NH_3
Select the correct option.

(1) a, c and d only

(2) b, c and d only

(3) a and b only

(4) c and d only

71. If the energy of ground state of hydrogen atom is –13.6 eV, then the energy of 1st excited state of He^+ ion in eV will be

(1) –6.04 eV

(2) –13.6 eV

(3) –19.62 eV

(4) –27.2 eV

72. Match **List-I** with **List-II**

List-I		List-II
A. XeO_3	(I)	sp^3d; linear
B. XeF_2	(II)	sp^3; pyramidal
C. $XeOF_4$	(III)	sp^3d^3; distorted octahedral
D. XeF_6	(IV)	sp^3d^2; square pyramidal

Choose the **correct** answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-IV, B-II, C-III, D-I

(3) A-IV, B-II, C-I, D-III

(4) A-II, B-I, C-IV, D-III

CC-044

Identify the **incorrect** statement for PCl_5 from the following.

(1) Orbitals of phosphorus are assumed to undergo sp^3d hybridization

(2) PCl_5 has two axial bonds stronger than three equational bonds

(3) The three equatorial bonds of PCl_5 lie in a plane

(4) The geometry of PCl_5 is trigonal bipyramidal

74. Which of the given molecules has the highest percentage of *d*-character in the hybrid orbital of central atom?

(1) SF_6

(2) XeF_6

(3) XeF_4

(4) BrF_5

75. Given below are two statements

Statement I: B_2 molecule has single pi bond.

Statement II: To convert N_2 to N_2^+, the removal of electron takes place from $\sigma 2p_z$ orbital.

In the light of above statements, choose the **correct** answer from the options given below.

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

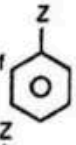
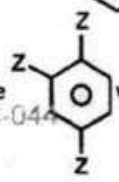
76. During formation of O_2 from O_2^-, electron is released from the orbital

(1) $\sigma 2p_z$

(2) $\sigma^* 2p_z$

(3) $\pi 2p_x$

(4) $\pi^* 2p_x$

77. If dipole moment of  is x D, then the dipole moment of molecule  will be

(1) 3x D

(2) 2x D

(3) x D

(4) $\frac{x}{2}$D

78. Match the species given in List-I with the number of unpaired electrons given in List-II

List-I	List-II
a. N	(i) 2
b. Cr^{2+}	(ii) 5
c. Mn^{2+}	(iii) 4
d. Ni	(iv) 3

The correct match is

(1) a(iv), b(iii), c(i), d(ii)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(iv), c(ii), d(i)

79. Which of the following statements are true regarding modern periodic table?

(a) *s*-block elements of the same group have similar valence shell electronic configuration.

(b) Metals comprise higher percentage than non-metals of the known elements.

(c) The physical and chemical properties of elements vary periodically with their atomic numbers.

Choose the **correct** answer from the options given below.

(1) (a) and (b) only

(2) (a) and (c) only

(3) (b) and (c) only

(4) (a), (b) and (c)

80. Which among the following processes is an endothermic process?

(1) $F(g) \rightarrow F^-(g)$

(2) $Br(g) \rightarrow Br^-(g)$

(3) $O^-(g) \rightarrow O^{2-}(g)$

(4) $O(g) \rightarrow O^-(g)$

81. Eka – aluminium and Eka - silicon in Mendeleev's periodic table respectively are

(1) Ga, Ta

(2) Ga, Sn

(3) Ge, Ga

(4) Ga, Ge

82. Match List-I with List-II.

List-I (Atomic number)	List-II (IUPAC Official Name)
a. 101	(i) Mendelevium
b. 103	(ii) Lawrencium
c. 102	(iii) Seaborgium
d. 106	(iv) Nobelium

Choose the correct option.

(1) a(i), b(ii), c(iv), d(iii)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(i), b(iii), c(iv), d(ii)

CC-044

83. In the Modern Periodic Table, the element with atomic number 33 belongs to

(1) 5th period and 15th group

(2) 3rd period and 13th group

(3) 4th period and 13th group

(4) 4th period and 15th group

84. Match list-I with list-II

	List-I		List-II
a.	CaO	(i)	Acidic
b.	As_2O_3	(ii)	Basic
c.	N_2O	(iii)	Amphoteric
d.	NO_2	(iv)	Neutral

The correct match is

(1) a-(ii), b-(iii), c-(i), d-(iv)

(2) a-(ii), b-(iii), c-(iv), d-(i)

(3) a-(iii), b-(ii), c-(iv), d-(i)

(4) a-(iv), b-(iii), c-(ii), d-(i)

85. Number of subshells present in 5th shell is

(1) 11

(2) 4

(3) 5

(4) 25

86. Given below are two statements

Statement (I): Sulphur has the most negative electron gain enthalpy among all the group 16 elements.

Statement (II): Fluorine is a stronger oxidising agent than chlorine.

In the light of above statements, choose the most appropriate option from the options given below.

(1) Both statement (I) and statement (II) are correct.

(2) Both statement (I) and statement (II) are incorrect.

(3) Statement (I) is correct but statement (II) is incorrect.

(4) Statement (I) is incorrect but statement (II) is correct.

87. Lithium is diagonally related with

(1) Be

(2) Mg

(3) Al

(4) Si

88. An atom with which of the following electronic configuration has the highest ionization enthalpy?

(1) $1s^2 2s^2 2p^1$

(2) $1s^2 2s^2$

(3) $1s^2 2s^2 2p^3$

(4) $1s^2 2s^2 2p^4$

89. Correct order of first ionization enthalpies is

(1) Li < Be < B < C

(2) Li < B < Be < C

(3) Li < C< Be < B

(4) Li < B < C < Be

90. Consider the following statements.

(i) Bromine is more electronegative as compared to sulphur (on Pauling scale).

(ii) Both Barium and Boron are main group elements.

(iii) Tin and lead are metals of *p*-block.

(iv) The number of unpaired electrons in *d* subshell of iron is 6.

The incorrect statements is/are

(1) (i) and (iv) only

(2) (iv) only

(3) (ii) only

(4) (i), (ii), (iii) and (iv)

BOTANY

91. Read the following Assertion (A) and Reason (R) and select the **correct** option.

Assertion (A) : Golgi apparatus remains in close association with the endoplasmic reticulum.

Reason (R) : Materials to be packaged in the form of vesicles from the ER fuse with cis face of the Golgi apparatus and move towards the maturing face.

(1) Only (A) is true but (R) is false

(2) Both (A) and (R) are false

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is not the correct explanation of (A)

Identify the following statements as true (T) or false (F) and accordingly select the **correct** option.
a. In prokaryotes mesosomes help in DNA replication and in respiration
b. Eukaryotic cells have a variety of complex locomotory and cytoskeletal structures
c. The middle lamella is a layer made up of magnesium pectate only
d. Plasmodesmata connects the cytoplasm of neighbouring cells

	a	b	c	d
(1)	T	T	T	T
(2)	T	F	T	F
(3)	F	F	T	T
(4)	T	T	F	T

CC-044 CC-044 CC-044 CC-044 CC-044 CC-044 CC-044

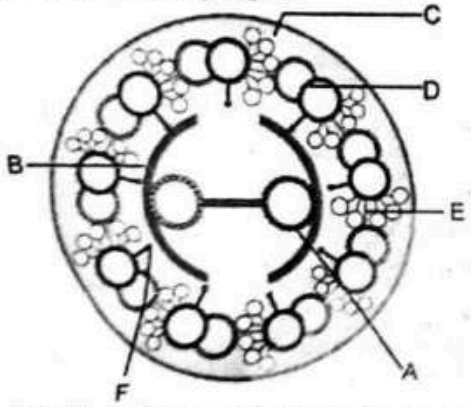
(1) (1)

(2) (2)

(3) (3)

(4) (4)

93. Observe the following diagram:



Select the option that **correctly** *depicts about the figure.*

(1) *Label A is composed of microfilaments*

(2) *The core in this structure is proteinaceous and called the hub.*

(3) *Label C represents central sheath*

(4) *This structure has 9 radial spokes and inter- doublet linkers*

94. Choose the **incorrectly** matched pair.

(1) Schwann – Presence of cell wall is unique character of the plant cells

(2) Schleiden – Reported that cells had a thin outer layer *i.e.* plasma membrane

(3) Rudolf Virchow – Explained that cells divide and new cells are formed from pre-existing cells.

(4) Schwann – Proposed the hypothesis that the bodies of animals and plants are composed of cells and products of cells

95. Satellite in chromosomes is

(1) Seen in telocentric chromosomes only

(2) A fragment beyond the secondary constriction

(3) Formed by primary constriction

(4) Devoid of DNA

96. The fluid nature of the membrane helps to perform functions like
(a) Cell growth & cell division
(b) Formation of intercellular junctions
(c) Endocytosis

(1) Only (c)

(2) (b) & (c) only

(3) (a) & (c) only

(4) (a), (b) & (c)

97. Fimbriae

(1) Are bristle-like structures found in all bacteria

(2) Are known to help to attach some bacteria to rocks in streams and also to the host tissues

(3) Help bacteria in process of genetic recombination

(4) Help in movement of bacteria

98. Select the **incorrect** option w.r.t. plant cell wall

(1) Secondary wall is formed towards inner side (towards membrane) of the cell

(2) Consists of cellulose, hemicellulose, pectins and proteins

(3) Primary cell wall gradually diminishes as the cell matures

(4) It is not a barrier to undesirable macromolecules as it is freely permeable

99. In human beings, the membrane of erythrocytes has approximately __(i)__% of proteins and __(ii)__% of lipids.
Here (i) and (ii) respectively are

(1) 40 and 52

(2) 52 and 40

(3) 40 and 60

(4) 60 and 40

100. Contractile vacuole in *Amoeba* helps in

(1) Protein synthesis

(2) Respiration

(3) Digestion

(4) Osmoregulation and excretion

101. Match the structures given in column-I with their functions/explanations given in column-II, and select the most appropriate answer.

	Column I		Column II
a.	Inclusion bodies	(i)	Increases surface area and enzymatic content of plasma membrane
b.	Plasmid DNA	(ii)	Several ribosomes attached to a single mRNA
c.	Polysome	(iii)	Reserve materials in the cytoplasm of prokaryotes
d.	Mesosome	(iv)	Confers certain unique phenotypic characters to bacteria

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iii), b(ii), c(iv), d(i)

102. Select the **incorrect** statement w.r.t. lysosome

(1) Part of endomembrane system

(2) Lysosomal enzymes do not digest nucleic acids

(3) Formed by packaging process in golgi apparatus

(4) Rich in hydrolytic enzymes

103. A spherical structure, which is the site of r-RNA synthesis, is

(1) Glyoxysome

(2) Nucleolus

(3) Perinuclear space

(4) Ribosome

104. The type of leucoplast which stores fat is

(1) Amyloplast

(2) Elaioplast

(3) Aleuroplast

(4) Chromoplast

105. Which of the given cell organelles is common to both prokaryotic and eukaryotic cell?

(1) True vacuole

(2) Ribosome

(3) Lysosome

(4) Golgi complex

106. Proteinaceous structure which maintains the shape of a cell, is

(1) Cell wall

(2) Cytoplasm

(3) Cytoskeleton

(4) Vacuole

107. Mitotic apparatus is formed of

(1) Two centrosomes with chromatids

(2) Two asters together with spindle fibres

(3) Two chromatids with acrosome

(4) One primary constriction with two chromatids

108. Interphase

(1) Involves DNA replication

(2) Constitute less than 50% duration of the cell cycle

(3) Is the phase of actual cell division

(4) Does not include RNA formation

109. Arrange the following events of meiosis I in the **correct** sequence of their occurrence

a. Terminalisation of chiasmata

b. Synapsis

c. Crossing over

d. Appearance of X-shaped structure

(1) $d \rightarrow c \rightarrow b \rightarrow a$

(2) $b \rightarrow c \rightarrow d \rightarrow a$

(3) $a \rightarrow b \rightarrow c \rightarrow d$

(4) $d \rightarrow a \rightarrow b \rightarrow c$

110. During which phase condensation of chromatin material takes place?

(1) Metaphase

(2) Anaphase

(3) Prophase

(4) Telophase

111. In animal cells DNA replication and centriole duplication respectively occurs in

(1) Nucleus and cytoplasm

(2) Cytoplasm and nucleus

(3) Cytoplasm and ER

(4) Golgi body and nucleus

112. Anaphase I of meiosis is different from anaphase of mitosis, as it shows

(1) Alignment of chromosomes at the equator

(2) Separation of homologous chromosomes towards the opposite poles

(3) Separation of sister chromatids towards the opposite poles

(4) Attachment of spindle fibres from opposite poles to the kinetochores of sister chromatids

CC-044

A stage of meiosis is shown in the diagram. Identify the stage and its appropriate characteristic?



(1) Anaphase of mitosis – Chromatids move towards the opposite pole

(2) Anaphase I of meiosis – Chromosomes start moving away

(3) Metaphase of mitosis – All the chromosomes lie at the equator

(4) Metaphase I of meiosis – The bivalent chromosomes align on the equatorial plate.

114. The phase that marks the end of cell division is concerned with

(1) Dissolution of the nucleus

(2) Division of the cytoplasm

(3) Untangling of the mitotic chromosome

(4) Splitting of centromere

115. In oocytes of some vertebrates, _____ lasts for months or years.
Choose the **correct** option to fill in the blank.

(1) Leptotene

(2) Pachytene

(3) Diplotene

(4) Diakinesis

116. All of the following are significances of mitosis, **except**

(1) Maintenance of cell size

(2) Healing and regeneration

(3) Growth of multicellular organisms

(4) Introduction of variations

117. Read the following statements A and B and choose the **correct** option.
Statement A : The number of similar characteristics goes on increasing from species to kingdom.
Statement B : Higher the category, greater is the difficulty of determining the relationship to other taxa at same level.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

118. State true (T) or false (F) and select the **correct** option.
(a) Genera are aggregates of closely related species.
(b) Families are characterised on the basis of both vegetative and reproductive features of plant species.
(c) Convolvulaceae and Solanaceae are included in the order Poales, mainly based on floral characters.
(d) The word systematics is derived from the Latin word 'systema'.

	a	b	c	d
(1)	T	T	F	F
(2)	T	T	F	T
(3)	F	T	F	T
(4)	F	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

119. Systematics is different from taxonomy as it takes into account

(1) Characterisation

(2) Identification

(3) Nomenclature

(4) Evolutionary relationships

120. Choose the **incorrect** option w.r.t. wheat.

(1) Genus – *Triticum*

(2) Family – Poaceae

(3) Order – Polymoniales

(4) Division – Angiospermae

121. Select the **correct** set of statements for the virus first crystallised by Stanley.
(a) It has either single or double stranded DNA as its genetic material.
(b) Subunits of their protein coat are arranged helically
(c) It is an obligate plant parasite.
(d) First discovered by M.W. Beijerinck.

(1) Only (b) and (c)

(2) Only (a) and (b)

(3) All except (d)

(4) All (a), (b), (c) and (d)

122. On the basis of mode of nutrition which of the following are most abundant in nature?

(1) Heterotrophic bacteria

(2) Photoautotrophic bacteria

(3) Symbiotic bacteria

(4) Chemoautotrophic bacteria

123. Prions are

(1) Larger than viruses

(2) Abnormally folded genetic material

(3) Causal agent of mad cow disease

(4) Devoid of proteins

124. Dikaryophase is **not** seen in

(1) *Aspergillus*

(2) *Puccinia*

(3) *Rhizopus*

(4) *Agaricus*

125. Chief producers in the ocean

(1) Are multicellular

(2) Have cell wall impregnated with silica

(3) Do not reproduce sexually

(4) Have flagella in vegetative phase

126. Read the given statements and choose the correct option.
Statement A: Protozoans are heterotrophs.
Statement B: Dinoflagellates release toxin into the sea water which is highly poisonous to some vertebrates.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

127. Match the following columns and select the **correct** option.

	Column I		Column II
a.	*Neurospora*	(i)	Edible ascocarps
b.	*Colletotrichum*	(ii)	Used extensively in biochemical and genetic work
c.	Morels	(iii)	Smut causing fungi
d.	*Ustilago*	(iv)	Only asexual phase is known

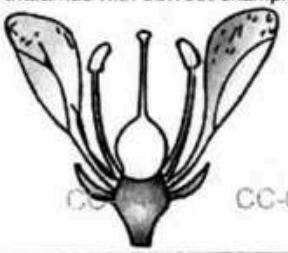
(1) a(ii), b(iv), c(iii), d(i)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iv), b(ii), c(i), d(iii)

128. Identify the figure w.r.t. the position of floral parts on thalamus with **correct** example.



	Position of floral parts on thalamus w.r.t. the ovary		Example
(1)	Epigynous	–	Plum
(2)	Perigynous	–	Rose
(3)	Hypogynous	–	Mustard
(4)	Perigynous	–	Brinjal

(1) (1)

(2) (2)

(3) (3)

(4) (4)

129. All are the main functions of root, **except**

(1) Absorption of water and minerals.

(2) Performs photosynthesis

(3) Synthesis of plant growth regulators

(4) Storing reserve food material

130. Identify the **wrongly** matched pair.

(1) Pinnately compound leaf – Neem

(2) Palmately compound leaf – Silk cotton

(3) Alternate phyllotaxy – China rose

(4) Opposite phyllotaxy – Mustard

(1) (1)

(2) (2)

(3) (3)

(4) (4)

131. Read the given statements (a-d)
a. Ovules develop into seeds after fertilization.
b. The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.
c. Nucellus remains persistent in some seeds and is called pericarp.
d. The plumule and radicle are enclosed in sheaths which are called as coleoptile and coleorhiza respectively.
In the light of above statements, select the **correct** option.

(1) All are correct except d

(2) Only b and c are correct

(3) Only a, b and d are correct

(4) Only c and d are correct

132. Select the **incorrect** statement about the aestivation

(1) When sepals or petals in a whorl just touch one another at the margin, without overlapping, it is said to be valvate

(2) If one margin of the appendage overlaps that of the next one, and so on, as in lady's finger, it is called twisted

(3) If the margins of sepals or petals overlap one another but not in any particular direction, it is called vexillary

(4) In bean flowers, there are five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel)

133. Root hairs develop from the region of

(1) Maturation

(2) Elongation

(3) Root cap

(4) Meristematic activity

Racemose inflorescence differs from cymose inflorescence as in the latter
(a) The shoot axis continues to grow indefinitely
(b) The main axis terminates into a flower
(c) The younger flowers are present towards the apex and the older flowers are present at the base
Select the **correct** one(s) from the options given below.

(1) (a) and (b) only

(2) (a) and (c) only

(3) (b) only

(4) (b) and (c) only

135. Which of the following characters is **not** related to family Solanaceae?

(1) It is a large family, commonly called as the 'potato family'

(2) Corolla is papilionaceous

(3) Seeds are many and endospermous

(4) Gynoecium is bilocular with swollen placenta and many ovules

CC-044 CC-044 CC-044 CC-044

ZOOLOGY

136. In cardiac muscle, which of the following at some fusion points mainly allow cells to contract as a unit?

(1) Adhering junction

(2) Neuromuscular junction

(3) Tight junction

(4) Intercalated discs

137. *Collagen fibres are **not** found in extracellular matrix of*

(1) Bone

(2) Adipose tissue

(3) Blood

(4) Areolar connective tissue

138. *Which of the following tissues exerts the greatest control over the body's responsiveness to changing conditions?*

(1) Muscle tissue

(2) Neural tissue

(3) Connective tissue

(4) Epithelial tissue

CC-044 CC-044 CC-044 CC-044 CC-044 CC-044

139. Select the incorrect statement.

(1) Areolar tissue is present beneath the skin.

(2) Adipose tissue contains fibroblasts and adipocytes.

(3) Dense irregular connective tissue contains collagen fibres.

(4) Loose connective tissue has a compact and dense matrix.

140. The tissue in which fibres and fibroblasts are compactly packed and which attaches skeletal muscles to bones possesses/also

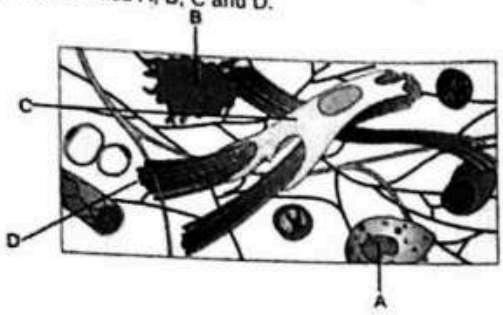
(1) Biomolecule which is the most abundant protein in the animal world

(2) Store excess of nutrients in the form of fats

(3) Acts as the site for production of blood cells

(4) Cells that are present in lacunae

141. Diagrammatic sketch of a certain connective tissue is given below. Select the option that **correctly** identifies the parts labelled A, B, C and D.



	A	B	C	D
(1)	Adipocyte	Mast cell	Collagen fibres	Macrophage
(2)	Collagen fibres	Osteocyte	Mast cell	Fibroblast
(3)	Macrophage	Mast cell	collagen fibres	Chondrocyte
(4)	Mast cell	Macrophage	Fibroblast	Collagen fibres

(1) (1)

(2) (2)

(3) (3)

(4) (4)

142. The major component which is present in plant cell wall does not give colour with iodine because it is/does

(1) A homopolymer

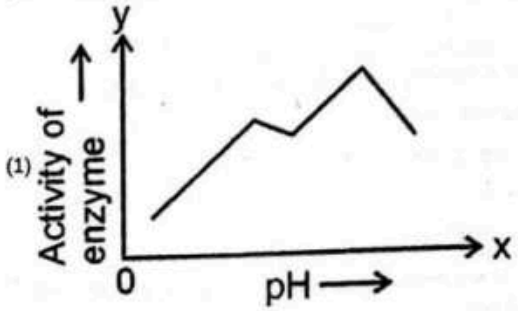
(2) A polysaccharide

(3) Not contain complex helices

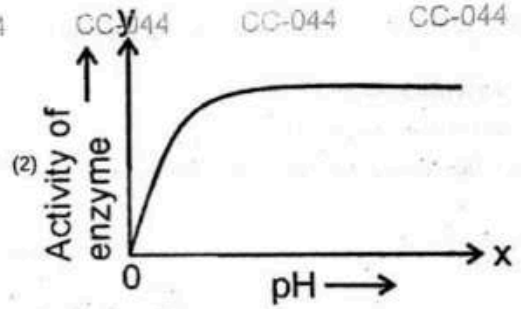
(4) Not have reducing end

143. Following graphs represent activity of enzyme versus pH. Choose the graph which **correctly** represents the effect of pH on the activity of enzyme.

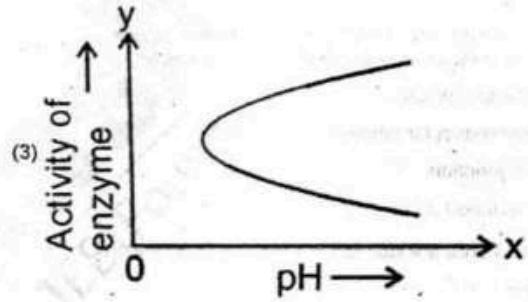
(1)



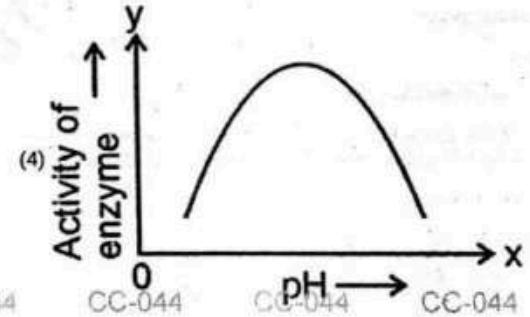
(2)



(3)



(4)



CC-044

Match column A and column B and choose the **correct** option.

	Column A		Column B
a.	Lectin	(i)	Morphine
b.	Alkaloids	(ii)	Abrin
c.	Toxin	(iii)	Vinblastin
d.	Drugs	(iv)	Concanavalin A
		(v)	Ricin

(1) a(iv), b(ii), c(v), d(iii)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(iii), b(iv), c(v), d(i)

(4) a(iii), b(v), c(ii), d(i)

145. Read the following statements.
A. In an exothermic reaction, the energy level of product is lower than that of substrate.
B. The difference in the average energy content of product from that of the transition state is called activation energy.
Choose the **correct** option.

(1) Statement A is correct but B is incorrect

(2) Statement B is correct but A is incorrect

(3) Both statements are incorrect

(4) Both statements are correct

146. **Assertion (A) :** Structures of amino acids are sensitive towards the solutions of different pH.
Reason (R) : $-NH_2$ and $-COOH$ groups present in the structure of an amino acid are ionizable in nature.
In the light of above statements, select the correct option.

(1) *(A) is true but (R) is false*

(2) *Both (A) and (R) are false*

(3) *Both (A) and (R) are true and (R) is the correct explanation of (A)*

(4) *Both (A) and (R) are true but (R) is not the correct explanation of (A)*

147. Choose the **incorrect** match.

(1)	Malleus, Incus, Stapes	–	Ear ossicles
(2)	Lacrimal bone, Nasal bone, Maxilla	–	Paired bones
(3)	Humerus, Radius, Ulna	–	Bones of forelimb
(4)	Muscular dystrophy	–	Disorder of skeletal system involving degeneration of bones

(1) (1)

(2) (2)

(3) (3)

(4) (4)

148. Removal of the co-factor from an enzyme will result in

(1) Increase of catalytic activity

(2) Initial decrease of catalytic activity followed by increase

(3) Loss of catalytic activity

(4) No effect on catalytic activity

149. Following is a representation of an enzymatic reaction

$$S-G+S' \xrightarrow{E} S+S'-G$$

Select the **correct** option which represents E for above reaction

(1) Lyase

(2) Ligase

(3) Transferase

(4) Isomerase

150. Read the given statements:
Statement-A : If the total amount of adenine in eukaryotic ds DNA is 30%, then total pyrimidine content will be 50%.
Statement-B : Nitrogenous bases do not form any part of the backbone in DNA.
Select the **correct** option.

(1) Only statement A is incorrect

(2) Only statement B is incorrect

(3) Both statements A and B are incorrect

(4) Both statements A and B are correct

151. The respiratory or exchange part of the respiratory system is formed by the

(1) Alveoli and their ducts

(2) Trachea upto secondary bronchi

(3) External nostrils upto trachea

(4) External nostrils upto terminal bronchioles

152. The reaction mentioned below is catalysed by enzyme 'X'.

$$CO_2 + H_2O \underset{}{\overset{\text{'X'}}{\rightleftharpoons}} H_2CO_3$$

Select the **correct** option w.r.t. 'X'.

(1) Catalase

(2) Peroxidase

(3) Carbonic anhydrase

(4) Peptidase

153. Under normal physiological condition, during the formation of oxyhaemoglobin, about 90% saturation of Hb with O_2 occurs when the pO_2 is about

(1) 40 mm Hg

(2) 60 mm Hg

(3) 55 mm Hg

(4) 45 mm Hg

154. Total volume of air accommodated in the lungs at the end of a forced inspiration can be represented as

(1) RV + ERV + TV + IRV

(2) TV + ERV

(3) IC + EC + RV

(4) IC + EC – TV

155. Diffusion membrane in lungs comprises:

(1) Thin columnar epithelium of alveoli, endothelium of pulmonary capillaries and basement membrane

(2) Ciliated squamous epithelium of alveoli, endothelium of pulmonary capillaries and basement substance between them

(3) Thin squamous epithelium of alveoli, endothelium of alveolar capillaries and basement substance between them

(4) Ciliated columnar epithelium of alveoli, endothelium of alveolar capillaries and basement membrane between them

156. Select the **correct** match.

(1)	Branchial respiration	–	Adult *Rana tigrina*
(2)	Cutaneous respiration	–	*Pheretima*
(3)	Pulmonary respiration	–	Prawn
(4)	Cutaneous respiration	–	Human

(1) (1)

(2) (2)

(3) (3)

(4) (4)

157. Select the correct option to complete the analogy.
pO_2 in oxygenated blood: 95 mmHg :: _____ : 45 mmHg

(1) pO_2 in alveoli

(2) pCO_2 in deoxygenated blood

(3) pO_2 in deoxygenated blood

(4) pCO_2 in alveoli

158. Match column I with column II w.r.t formed elements and their percentage in total WBCs in a normal adult human.

	Column I		Column II
i.	Neutrophils	(a)	2-3 %
ii.	Basophils	(b)	20-25 %
iii.	Eosinophils	(c)	60-65 %
iv.	Lymphocytes	(d)	0.5-1 %

Select the **correct** option.

(1) i(c), ii(d), iii(a), iv(b)

(2) i(a), ii(b), iii(d), iv(c)

(3) i(d), ii(b), iii(a), iv(c)

(4) i(d), ii(b), iii(c), iv(a)

159. Blood from intestine is carried to the liver via _____ before it is returned to systemic circulation. The **correct** option which fills the blank is

(1) Hepatic vein

(2) Hepatic portal vein

(3) Hepatic artery

(4) Inferior mesenteric artery

160. A special connection of blood vessels, present in humans, exclusively for the circulation of blood to and from the cardiac muscles is called

(1) Coronary system

(2) Renal portal system

(3) Hepatic portal system

(4) Lymphatic system

161. Choose the correct statement w.r.t the cardiac cycle of a healthy individual under normal physiological conditions.

(1) The volume of blood pumped out by the right and left ventricles is same.

(2) The volume of blood pumped out by the right and left ventricles is significantly different.

(3) Complete filling of ventricles occur during joint diastole.

(4) Amount of blood pumped out in each cardiac cycle is called cardiac output.

162. The pulmonary circulation starts by the pumping of $\frac{A}{B}$ blood by the right ventricle which is carried to the $\frac{B}{A}$.

Choose the **correct** option that fills the blanks A and B respectively.

(1) Oxygenated; Heart

(2) Oxygenated; Lungs

(3) Deoxygenated; Lungs

(4) Deoxygenated; Heart

163. Heart disorder in which cardiac muscles are suddenly damaged by inadequate supply of blood is

(1) Heart failure

(2) Cardiac arrest

(3) Heart attack

(4) Atherosclerosis

164. Consider the given features w.r.t human circulatory system.
(a) Possesses the thickest muscular wall
(b) Walls are innervated by Purkinje fibres
(c) Pumps oxygenated blood
The above mentioned features are true for the chamber of heart which

(1) Has SA node in its wall

(2) Receives blood directly from pulmonary artery

(3) Pumps blood to lungs *via* pulmonary vein

(4) Pumps blood to the Aorta with maximum blood pressure

165. Read the following statements w.r.t. the Henle's loop in humans:
(A) It plays a significant role in the maintenance of low osmolarity of the medullary interstitial fluid.
(B) Its descending limb is permeable to electrolytes but impermeable to water.
(C) Reabsorption is maximum in its ascending limb.
(D) Its ascending limb allows transport of electrolytes actively or passively.
Which of the above given statements are **not** true?

(1) (A) and (B) only

(2) (C) and (D) only

(3) (A), (B) and (D) only

(4) (A), (B) and (C) only

166. Under normal physiological conditions, all of the following hormones increase the blood pressure in a healthy humans, **except**

(1) Vasopressin

(2) Aldosterone

(3) Angiotensin II

(4) ANF

167. Which of the following is **not** a true statement about haemodialysis?

(1) Blood is taken out from an artery of the patient, cooled to 0°C and mixed with antiheparin

(2) The blood flows inside the cellophane tubes which are impermeable to plasma proteins

(3) Dialysing fluid does not contain nitrogenous wastes

(4) Blood coming out of artificial kidney is warmed to body temperature, mixed with anti-heparin and returned to patient's vein

168. Podocytes are particularly found in

(1) Glomerular capillaries

(2) Neck region of nephron

(3) Inner wall of Bowman's capsule

(4) Outer wall of Bowman's capsule

169. Arrange the following organisms in the correct order w.r.t. the level of toxicity of the chief nitrogenous wastes that they excrete.
A. Bony fishes
B. Aquatic amphibians
C. Terrestrial amphibians
D. Reptiles
Select the **correct** option.

(1) A < B < C = D

(2) C > D = A > B

(3) A = B > C > D

(4) D > B > C = A

170. If we compare the volume of the filtrate formed per day with that of the urine released per day in a healthy adult human under normal physiological conditions, then the ratio obtained will be most likely

(1) 99 : 1

(2) 100 : 1

(3) 120 : 1

(4) 200 : 1

171. If PCT part of nephrons is removed from mammalian kidney then

(1) Urine will become more diluted

(2) Urine will become more concentrated

(3) No change will occur in concentration of urine

(4) Urine formation will not occur

172. Which of the following parts of scapula articulates with head of humerus?

(1) Spine

(2) Glenoid cavity

(3) Acromion process

(4) Glenoid cavity and acromion process

173. Consider the following statements w.r.t. humans.
(a) Patella and sternum are the parts of appendicular skeleton.
(b) Scapula is situated in the dorsal part of thorax between the second and seventh rib.
(c) Thoracic vertebrae participate in the formation of ribcage.
Choose the **correct** option.

(1) (a), (b) and (c) are correct

(2) Only (a) and (b) are correct

(3) Only (b) and (c) are correct

(4) Only (c) is correct

174. **Assertion (A):** In competitive inhibition, both substrate and inhibitor simultaneously bind to the same active site of the enzyme.
Reason (R): Competitive inhibitors change both the K_m and V_{max} values of the catalysed reaction.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is correct explanation of the (A).

(2) Both (A) and (R) are true but (R) is not correct explanation of the (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

CC-044

175. Which of the following is not a correct point of difference between a red muscle fibre and a white muscle fibre?

	Red muscle fibre	White muscle fibre
(1)	High myoglobin content	Very low myoglobin content
(2)	High amount of sarcoplasmic reticulum	Low amount of sarcoplasmic reticulum
(3)	More number of mitochondria	Less number of mitochondria
(4)	Reddish in appearance	Pale or whitish in appearance

(1) (1)

(2) (2)

(3) (3)

(4) (4)

176. A single, U-shaped bone present at the base of the buccal cavity in humans is

(1) Hyoid

(2) Mandible

(3) Sphenoid

(4) Occipital

177. Read the given statements:

(a) The number of cervical vertebrae are seven in all mammals.

(b) Each vertebra of human has a central solid portion called the neural canal.

(c) Humans and frogs have dicondylic skull.

(d) Sternum is a triangular flat bone on the ventral midline of thorax.

How many of them is/are correct?

(1) Four

(2) Three

(3) Two

(4) One

178. The cross bridge formation during muscle contraction occurs just after

(1) Neurotransmitters are released at the neuromuscular junction

(2) Release of Ca^{2+} in the sarcoplasm

(3) Binding of Ca^{2+} to a subunit of troponin

(4) Binding of head of myosin with actin by utilizing the energy from ADP hydrolysis

179. Each A has many parallely arranged B. Choose the option which fills the blanks A and B correctly.

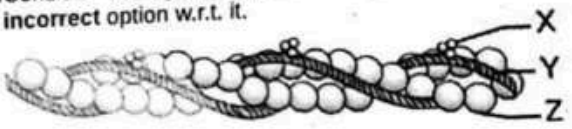
(1) **A**-Myofibril, **B**-Neurofibrils

(2) **A**-Myofibril, **B**-Muscle fibres

(3) **A**-Muscle fibre, **B**-Neurofibrils

(4) **A**-Muscle fibre, **B**-Myofibrils

180. Consider the given labelled diagram and select the **incorrect** option w.r.t. it.



(1) A subunit of 'X' masks the active binding sites for myosin on the thin filament

(2) Each 'Z' is a polymer of monomeric G-actins.

(3) 'Y' is a homopolymer of glucose.

(4) 'X' is a trimeric protein.

CC-044